



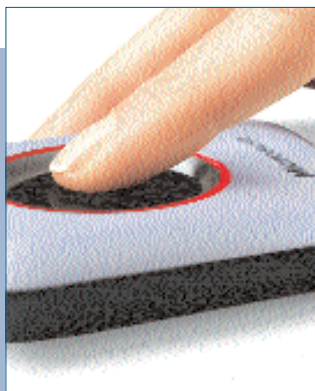
Doctors need not travel far at the risk 10 patients' lives to operate on one patient

paper) and read it on your handheld device. Technologies that are enabling foldable OLEDs (Organic Light Emitting Diodes) can make the experience even richer. You just plug in the sheet of screen to your computer (or phone) and download the day's newspaper. Navigation keys placed right on the sheet will help you read through the entire newspaper.

Of course, these services would be paid for, but they are at your convenience. Like newspapers, books and even movies or music could then be downloaded to such devices and accessed at your convenience. That, in the future, would be the true operative word. Convenience.

Education

Videoconferencing and virtual classrooms already exist, and this is one of the best marriages of technology, the Internet and a necessity that we have yet seen. In areas where schools or colleges are inaccessible, but telephone lines are available, virtual classrooms or teaching via videoconferencing can prove to be interactive and almost as good as the real thing. Using tools such as instant messaging and e-mail can further boost practicability. Reliance Infocomm, for example, offers good connectivity, and could be perfect for this purpose.



Distance learning (left) and fingerprint identity (right) are two areas where convergence of the old and the new can facilitate things a great deal

Good old Doordarshan has been telecasting the UGC classrooms on its National Network for a while and it the programme has been popular. Even without videoconferencing, education can reach a large audience thanks to technology. Just VCD players—or even VCRs for that matter—can aid in this. It's just a question of the drive being there.

Agriculture And Farming

Picture this: a farmer has a huge crop and it has been hit by a bug, but he has no idea which one it is... no one around seems to know. He doesn't have enough time to go to the big city and get it analysed and then treated. But somewhere in the annals of the World Wide Web lie the answers. Problem is, he cannot access it since he does not know English. But, thanks to the one computer- and Internet-savvy person in the village, he finds the answer to his problem and saves his year's crop.

Taking it further, from the consumer point of view, an animal being reared for meat could be tracked from birth. Information about its parents,

medical history, diet history and the way it was killed could be made available just through the bar code on the final package. Buyers could just scan this using their cell phone and get the information on their WAP-enabled cell phones before making a buying decision. Why not?

The camera phone could work well for farmers as well. In the first scenario, instead of the farmer going on to the Internet, he could just take a photo of the bug and MMS it to the nearest authority—government or private—and get personalised information and help faster than any other means. Moreover, this information could then be logged for the future reference of other farmers.

Healthcare

One of the major fields of convergence of basic, consumer-friendly technology could be healthcare. The worst time for a patient is after he suffers his injury or ailment and before medical help arrives. Using cameras or camera phones, in such situations could help address this better.

In fact, in an ideal world, a doctor need not leave his chamber again unless absolutely necessary. It's akin to telecommuting for doctors. Agreed, this may not be the most convenient way in all cases, but in some, it could definitely save doctors and patients a lot of time and grief.

In today's day and age, when operations are also happening via videoconferencing, nothing is fantasy. Experts, in particular, need not travel thousands of kilometres at the risk of 10 other patients' lives to operate on one patient. There are doctors who are competent enough to successfully operate with just the expert's guidance, and this is already reality. It's like TeleOperate.

Identity protection

Fingerprint identity or cornea scanning could combine the uniqueness of the human body and technology to create perhaps the most secure form of identity protection.

Contrary to what films may show us, there is a certain amount of surety this form of security lends us. A fingerprint is made of a series of ridges and furrows on the surface of the finger. The uniqueness of a fingerprint can be determined by the pattern of ridges and furrows as well as the minutiae points. Minutiae points are local ridge characteristics that occur at either a ridge bifurcation or a ridge ending. The other method is the correlation based method that is more practical since matching minutiae in low-quality scans could be difficult.

But where can it be implemented? Can colleges and schools have this to avoid cheating? Can offices secure data and maintain the secrecy of classified information? Can governments implement this to ensure that information is not stolen and misused? The first example may be a bit far-fetched, but the other two are certainly practical and have, in fact, been implemented in certain setups.

Prohibitive costs are the main hindrance, although devices from companies like Sony and Microsoft are aimed at the mass market.

Plus, the huge amount of data and the cumbersome nature of including new and dynamic data could prove restrictive. ■

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